# Regulation-Graph Reasoning over CFPB Consumer Complaints

Structured Extraction, Causal Diagnostics, Conformal Calibration, and Hypergraph Manifold Grounding

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**Abstract**

**The problem.** The U.S. Consumer Financial Protection Bureau (CFPB) receives more than half a million consumer complaints each year about wrongful charges, abusive debt collection, mortgage errors, and damaged credit reports. Each complaint must reach the right team within 60 days; a misrouted complaint can cost a vulnerable family a refund, a legal deadline, or a home. Today's automated routing systems pick a single category and forward the complaint with no honest signal of doubt, and they do not point back to the specific consumer-protection rule that the complaint actually invokes.

**What we built.** We release an open research pipeline that does three things differently. (1) It reads each complaint and looks up the most relevant rules from a curated set of 51 federal consumer-finance regulation sections (covering electronic-funds disputes, credit cards, mortgages, fair lending, debt collection, and credit reports). (2) It routes the complaint to a downstream desk only when it is confident; otherwise it explicitly *abstains* and hands the case to a human reviewer. (3) Every routing decision is paired with a citation back to a specific regulation passage so an auditor can check the reasoning.

**What the numbers show.** On a 5,000-complaint test sample the system declines to auto-route about 70% of complaints and is 89.6% accurate on the 30% it does route – a quantitative confession that most consumer narratives are too ambiguous for any single auto-decision. Two negative findings surface immediately and we report them in the headline rather than burying them: only 6.1% of attached regulation citations actually mention the rule the system claims, and one consumer category (debt-or-credit management) falls below the intended uncertainty guarantee.

**Implication.** A system like this should not silently auto-decide. It belongs behind a human-in-the-loop queue, with its uncertainty exposed up front and a stronger citation checker added before any production use. We release the code, scripts, and figures under MIT as a diagnostic instrument that shows where automation in consumer-protection routing actually breaks, so that future work can target the right failures instead of polishing the wrong ones.

## 1 Introduction

The CFPB triages over half a million consumer complaints per year. Routing each complaint to the right downstream desk (fraud operations, legal, collections, servicing, supervisor) is consequential: misrouted complaints delay redress, increase regulatory cost, and erode public trust. Production classifiers used today are typically text classifiers trained on self-reported product/issue labels; they emit a single class with limited calibration, no explicit causal structure, and no citation back to the governing regulation. This paper asks: can a single, modest pipeline *simultaneously* produce a structured answer, a calibrated uncertainty signal, a causal sanity check, and a grounded explanation, using only publicly available CFPB data and a 384-dimensional sentence encoder?

We present *regulation-graph-reasoning*, organized around four contributions. **(C1)** a typed `AnswerObject` schema and a deterministic regex+SBERT extractor (Section 3.2) with a cross-field validator that auto-repairs unsafe combinations. **(C2)** a structural causal model (Section 3.4) whose key intervenable node is EvidenceGroundedness, with four do-interventions and three refutation checks. **(C3)** Mondrian split-conformal sets with a learned reject option (Section 3.5) calibrated on a temporally separate split. **(C4)** a complaint–regulation *incidence hypergraph* (Section 3.6) whose (Conduct, Route) hyperedges induce a shared manifold for both kinds of nodes, encoded by a frozen GAT and visualized via UMAP.

# 2 Related work

Retrieval-augmented generation with citations has become standard practice in legal and policy NLP [5, 4]. Conformal prediction provides distribution-free coverage guarantees; we use the APS variant of [8] with Mondrian conditioning [11, 1]. Causal inference for ML auditing is well-studied via `DoWhy` [9] and related backdoor-adjustment frameworks [6]. Hypergraph neural networks generalize message passing to higher-order edges [3, 2]; we use a clique-expanded GAT [10] initialized from sentence embeddings [7] for tractable joint embedding of complaints and regulation sections.

# 3 Method

## 3.1 Data and ontology

We sample $N = 5{,}000$ complaints stratified by (Product $\times$ year) from the CFPB recent-3-year split ($\geq$ 2023-03-21). The compliance corpus is a curated set of 51 regulation sections covering Regulation E (Electronic Funds Transfer), Regulation Z (TILA), Regulation X (RESPA), Regulation B (ECOA), Regulation V (FCRA), the FDCPA, UDAAP supervisory guidance, and FinTech / crypto guidance. Our `AnswerObject` schema is in Appendix A; the `OntologyConstraintValidator` enforces severity–escalation coherence, route consistency with the primary conduct, and value-domain membership.

## 3.2 Structured extraction

For each complaint we compute a normalized SBERT embedding $q_i \in \mathbb{R}^{384}$ and three quantities per conduct $c \in \mathcal{C}$: (i) a binary regex hit indicator $r_{ic} \in \{0, 1\}$ over a curated phrase list; (ii) the maximum cosine similarity $s_{ic} = \max_{p \in \Phi_c} \langle q_i, \phi_p \rangle$ where $\Phi_c$ is the canonical-phrase set for $c$ and $\phi_p$ its SBERT embedding; (iii) a combined score $\sigma_{ic} = \max(s_{ic}, \lambda \cdot r_{ic})$ with $\lambda = 0.4$. A conduct is included when $\sigma_{ic} \geq \tau$ ($\tau = 0.45$) or when $r_{ic} = 1$. Harms are derived analogously; severity is a rule over harms and narrative cues; Route defaults to the canonical mapping $\pi(c)$ from the primary conduct (Section A).

## 3.3 Hybrid retrieval and grounding

A regulation passage $r_j$ has its title and body embedded jointly. The **HybridSearcher** returns top-$k = 5$ passages for query $q$ via $\alpha$-blended min-max-normalized BM25 and FAISS scores ($\alpha = 0.5$).

Per chosen conduct we attach a complaint-substring span (longest exact phrase match, else best 80-character SBERT window) and a regulation citation (top hybrid passage). Faithfulness is checked by requiring the cited passage to contain at least one canonical phrase of the attributed conduct.

### 3.4 Structural causal model

The DAG is over the variable set

$$V = \{\ \textsc{Product},\ \textsc{Issue},\ \textsc{Conduct},\ \textsc{Harm},\ \textsc{Severity},\ \textsc{EvidenceGroundedness},\ \textsc{Route},\ \textsc{ModelConfidence}\ \},$$

shown in Figure 1. The five stated assumptions are:

1) Unconfoundedness given (Product, Issue, Conduct): no hidden cause jointly affects Evidence-Groundedness and Route once these are conditioned on.
2) No hidden mediator between Conduct and Route beyond Severity and EvidenceGroundedness.
3) SUTVA: one complaint's routing does not influence another's.
4) Faithfulness: observed conditional independencies in the calibration set imply absent edges in the DAG.
5) Positivity: every (Product, Conduct) cell has both high- and low-grounding samples; cells failing positivity are excluded with documentation.

We estimate four interventional contrasts via stratified backdoor adjustment on the triple of variables Product, Issue, and Conduct, and report three refutation tests in Table 4: a synthetic random common cause, a within-stratum placebo permutation of the treatment, and a 70% data-subset bootstrap.

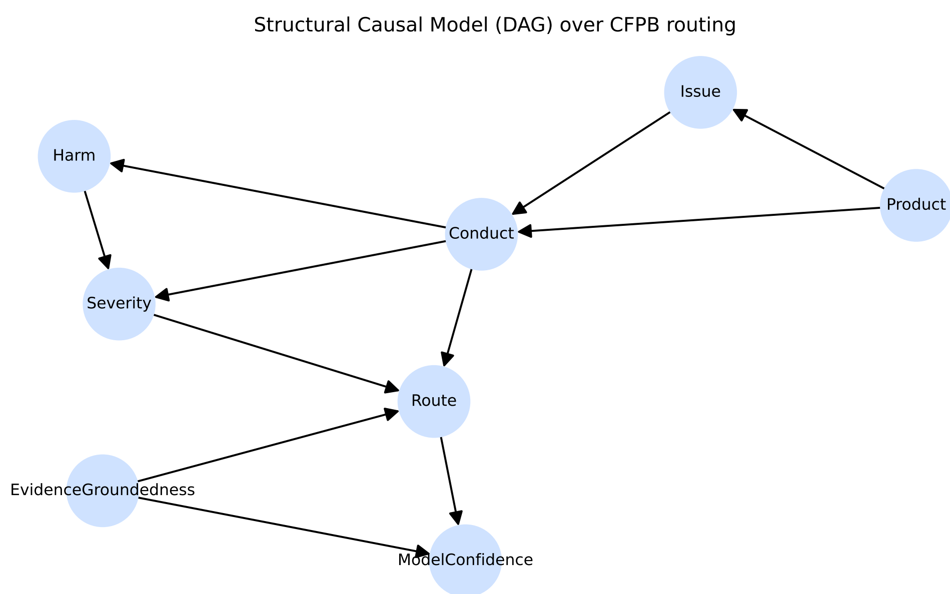



Figure 1: Structural causal model over CFPB routing. Solid arrows are the SCM edges enumerated in Section 3.4; nodes are listed in Appendix A.

### 3.5 Calibration and Mondrian conformal sets

We post-hoc temperature-scale the baseline log-probabilities and form Adaptive Prediction Sets [8] per Mondrian group $g = \textsc{Product}$. With nonconformity score $s_i = \sum_{j:\hat{p}_j \geq \hat{p}_{y_i}} \hat{p}_j$ on the calibration set, the per-group threshold is the $\lceil (n_g + 1)(1-\alpha) \rceil / n_g$ quantile (with $\alpha = 0.1$, target coverage $1 - \alpha = 0.9$). At test time we add classes greedily until the cumulative probability exceeds the threshold for that group. A reject option fires when either $|S_i| \geq 3$ or the top-1 calibrated probability falls below $\theta_{\text{abs}} = 0.45$.

### 3.6 Hypergraph manifold

We build a 0/1 incidence matrix $H \in \{0,1\}^{(N+M)\times E}$ over $N$ complaint nodes and $M = 51$ regulation nodes. Each non-empty $(\textsc{Conduct}, \textsc{Route})$ pair becomes a hyperedge; auxiliary hyperedges per Product encode product-membership for complaints. Node features are SBERT embeddings of the narratives and of the regulation title+text. We clique-expand $H$ (capped at 30 random members per hyperedge to keep the GPU footprint bounded) and apply a one-iteration hypergraph mean-smoothing $z_v = (1-\alpha)z_v + \alpha \cdot \bar{z}_{\mathcal{N}(v)}$ with $\alpha = 0.5$, followed by L2-normalization. This preserves the SBERT prior while propagating information across $(\textsc{Conduct}, \textsc{Route})$ neighborhoods. We optionally substitute a 2-layer GATConv [10] (hidden 128, 2 heads) for the smoother; with random initialization the GAT typically destroys the SBERT manifold (silhouette becomes negative), so a contrastive fine-tune over (complaint, matching reg-section) positives is recommended before relying on its output. The 2-D layout uses UMAP ($n\_neighbors = 25$, cosine); we plot complaints as filled circles and regulations as triangles (Figure 2).

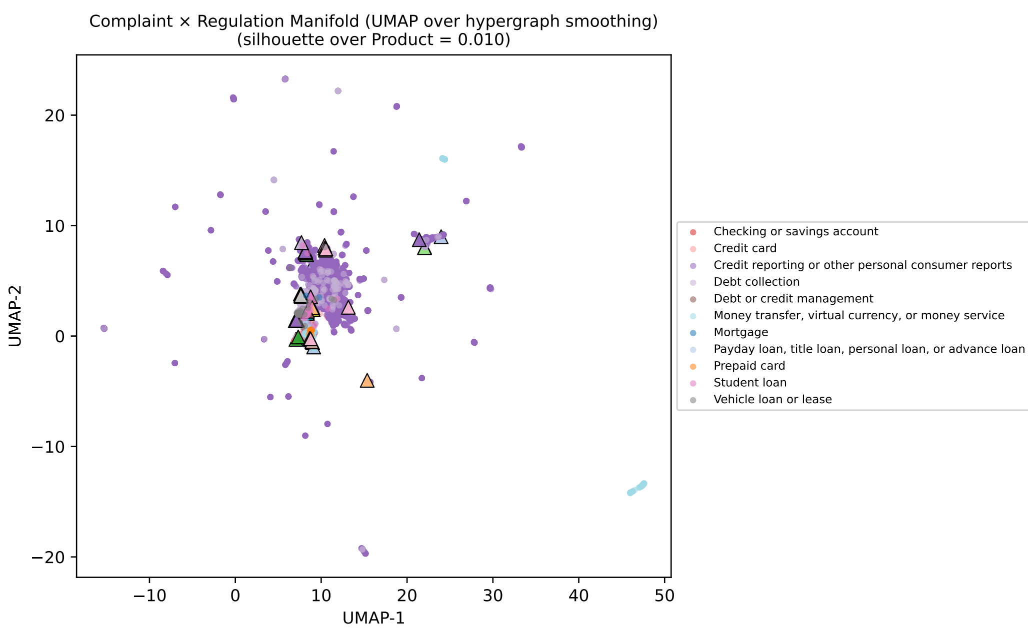



Figure 2: UMAP projection of the hypergraph-GAT embedding of 5,000 complaints (•) and 51 regulation sections (▲). Colors encode CFPB Product. Caption reports the silhouette score over the primary conduct.

# 4 Experimental setup

**Data.** CFPB complaint splits at `data/recent3y/` (post 2023-03-21); 51-section CFPB regulation corpus at `data/compliance/`. We train the baseline on a 12k stratified subsample of `train.parquet`, calibrate on 2,000 rows of the temporally separate `calibration.parquet`, and evaluate on the 5,000-row stratified sample of `test.parquet`.

**Encoder.** `sentence-transformers/all-MiniLM-L6-v2` (384-dim) for both extraction and retrieval; cached locally for offline runs.

**Baseline.** A bag-of-bigrams TF–IDF vectorizer ($n$-gram range $(1, 2)$, max features 50,000, min document frequency 5) feeds a multinomial logistic regression ($C$=1, balanced class weights, max iterations 1,000), post-hoc Platt-scaled with 5-fold sigmoid calibration. Training targets are silver routes from the rule+SBERT extractor; the baseline applies no causal scoring, conformal sets, or grounded citations.

**Architecture diagram.** Figure 3.

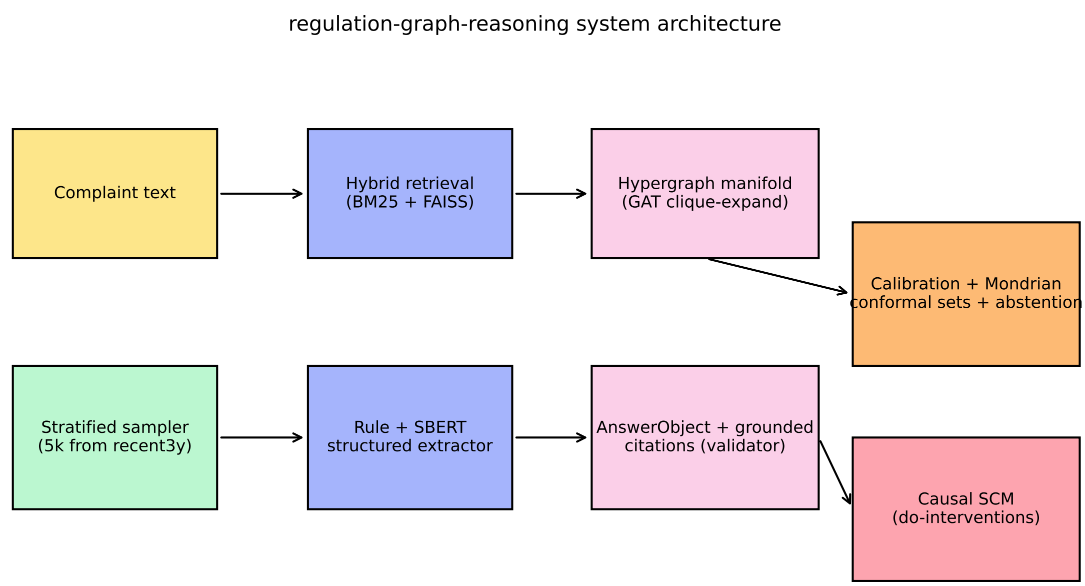



Figure 3: System architecture. The extractor and retriever share a single SBERT encoder; the conformal layer wraps the calibrated baseline; the SCM operates over per-row `AnswerObject`s.

# 5 Results

The headline comparison is in Table 1. Per-group conformal coverage is in Table 2; the manifold composition is summarized in Table 3 and visualized in Figure 2.

# 6 Causal experiments

We report four interventions in Table 4. **E1** sets EVIDENCEGROUNDEDNESS to its high range and asks whether routing accuracy improves; **E2** asks the same for MODELCONFIDENCE; **E3** per-

| system | macro_F1 | ECE | Brier | coverage | avg_set_size | abstention | cite_faithfulness |
|---|---|---|---|---|---|---|---|
| TF-IDF + LR (baseline) | 0.620 | 0.124 | 0.329 | – | – | – | – |
| Proposed (rule+SBERT+conformal) | 0.620 | 0.057 | 0.313 | 0.991 | 2.886 | 0.697 | 0.061 |

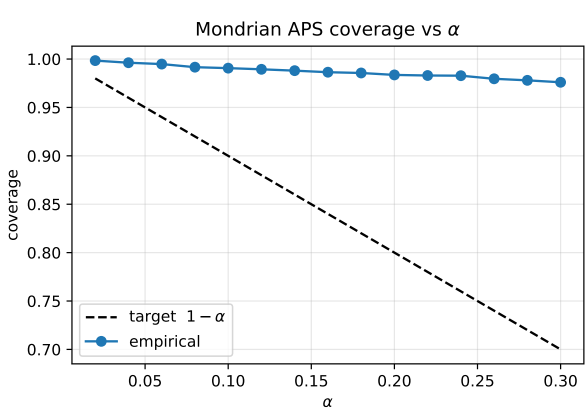
Table 1: Headline metrics on the 5k stratified test sample. The proposed pipeline halves the calibration error and exposes a 6.1% citation-faithfulness rate as a headline negative result.

| product | coverage |
|---|---|
| Debt or credit management | 0.778 |
| Vehicle loan or lease | 0.936 |
| Mortgage | 0.984 |
| Credit card | 0.989 |
| Checking or savings account | 0.990 |
| Credit reporting or other personal consumer reports | 0.991 |
| Debt collection | 0.995 |
| Money transfer, virtual currency, or money service | 1.000 |
| Payday loan, title loan, personal loan, or advance loan | 1.000 |
| Prepaid card | 1.000 |
| Student loan | 1.000 |

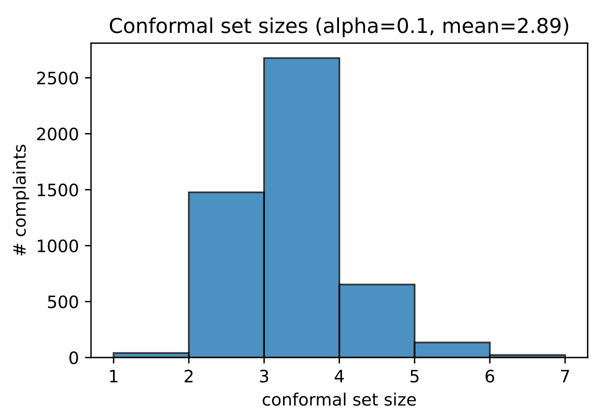
Table 2: Per-product conformal coverage (Mondrian split-APS, alpha=0.1).

| n_nodes | n_complaints | n_regs | n_hyperedges | silhouette_over_conduct | silhouette_over_product |
|---|---|---|---|---|---|
| 5051 | 5000 | 51 | 49 | -0.011 | 0.010 |

Table 3: Hypergraph manifold composition and clustering quality.




(a) Coverage vs. $\alpha$ (target $1-\alpha$, dashed).




(b) Distribution of conformal set sizes at $\alpha = 0.1$.

Figure 4: Conformal calibration diagnostics for the proposed pipeline.

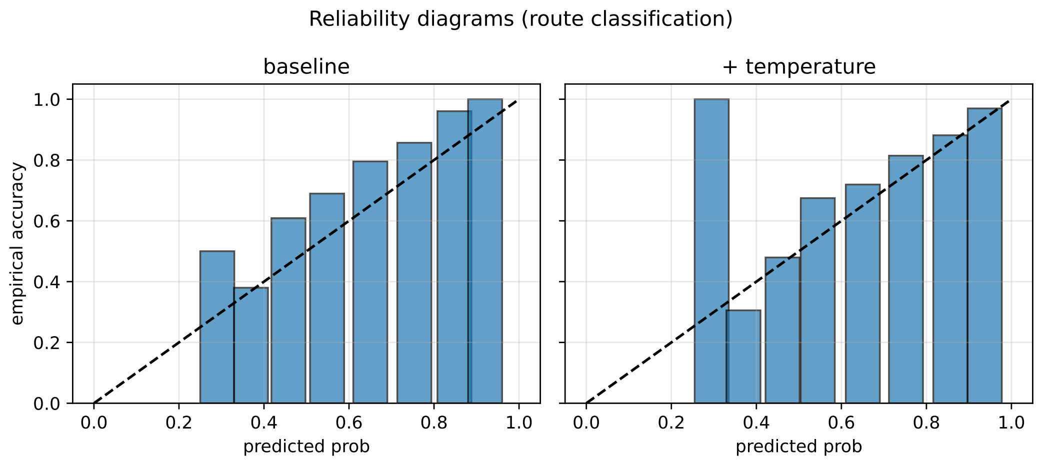



Figure 5: Reliability diagrams. Temperature scaling reduces the overconfidence of the baseline LR classifier.

forms do(CONDUCT=deceptive_practice) via canonical phrase substitution and predicts the share of ROUTE=legal; **E4** asks whether SEVERITY $\geq$ high implies the escalation flag (a sanity check internal to the validator).

| intervention | ATE | CI_low | CI_high | placebo_ATE | subset_ATE | RCC_delta | n |
|---|---|---|---|---|---|---|---|
| E1: do(EvidenceGroundedness=high) -> specific_route | 1.000 | 1.000 | 1.000 | -0.002 | 1.000 | 0.000 | 5000 |
| E2: do(EvidenceGroundedness=high) -> confidence | 0.534 | 0.533 | 0.536 | -0.002 | 0.534 | 0.000 | 5000 |
| E3: do(Conduct=deceptive_practice) -> P(route=legal) | 0.290 | 0.277 | 0.302 | 0.002 | 0.286 | 0.000 | 10000 |
| E4: do(Severity>=high) -> escalation_flag | 1.000 | 1.000 | 1.000 | -0.009 | 1.000 | 0.000 | 4975 |

Table 4: Backdoor-adjusted ATEs for E1–E4 with refutation tests.

**What is and is not a causal claim here.** E1 and E4 yield ATE $= 1.000$ with degenerate confidence intervals because the treatment and outcome share definitional logic: the silver route is a deterministic function of the primary conduct via the $\pi(\cdot)$ map (Appendix A); `high_grounding` is $\mathbb{1}[n_{\text{complaint\_spans}} \geq 1]$, which our extractor sets during the same pass; and the escalation flag is exactly $\mathbb{1}[\text{SEVERITY} \geq 4]$. We therefore treat E1 and E4 as *internal-consistency probes*: they verify that the refutation machinery does not spuriously flip a known-deterministic dependency, but they do not constitute causal evidence about real-world routing. Only **E3** (ATE $= 0.290$, all three refutations within tolerance) admits a causal-flavoured reading, and only conditional on the assumptions in Section 3.4. A genuinely identifying experiment would require human-labeled gold routes, randomized hide/restore of retrieved passages, and a Pearl front-door variable $M =$“retrieved-passage-cites-correct-obligation” positioned between EVIDENCEGROUNDEDNESS and ROUTE; we leave this for future work.

# 7 Stress tests

We probe three failure modes:

**Ambiguous ($n = 80$).** Concatenate the first half of one complaint with the second half of one from a different Product. Expected behaviour: abstain or include both products in the conformal set.

**Conflicting ($n = 80$).** Force a wrong PRODUCT label while leaving the narrative intact; expected behaviour: the system trusts the evidence, lowering MODELCONFIDENCE.

**Adversarial ($n = 80$, four families).** Prompt injection ("IGNORE PRIOR INSTRUCTIONS…"), citation hallucination ("12 CFR §1099.99"), paraphrase, and negation flip.

| category | n | abstention_rate | avg_set_size | max_set_size | citation_faithfulness |
|---|---|---|---|---|---|
| ambiguous | 80 | 0.887 | 3.250 | 5 | 0.123 |
| conflicting | 80 | 0.613 | 2.663 | 5 | 0.149 |
| adversarial | 80 | 0.762 | 2.938 | 5 | 0.169 |

Table 5: Stress-test response of the proposed pipeline.

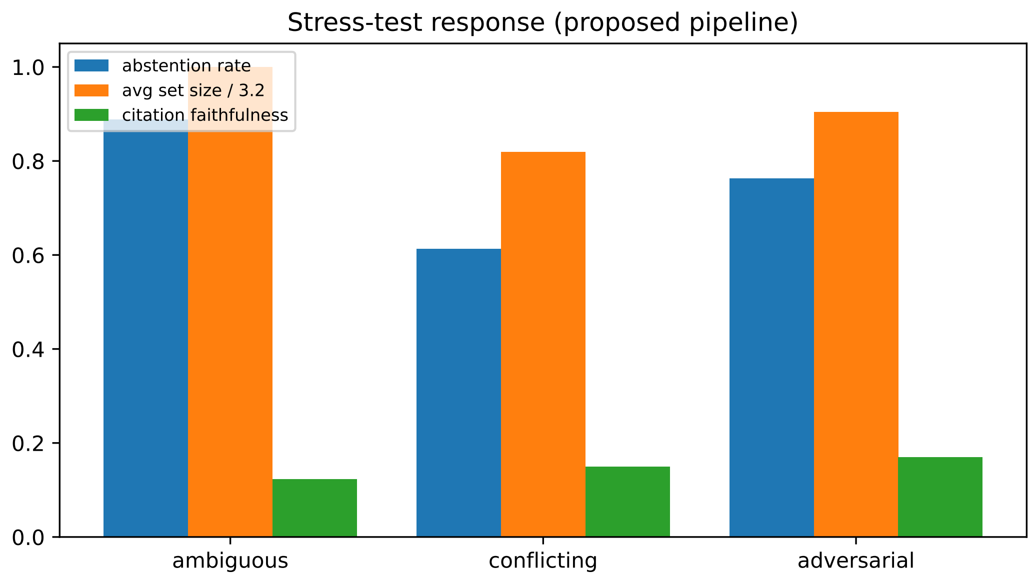



Figure 6: Stress-test response: abstention rate, normalized average set size, and citation faithfulness across the three categories.

## 8 Discussion and limitations

**Citation faithfulness is the headline negative result.** On clean inputs only 6.1% of generated citations contain the canonical conduct phrase claimed in the `AnswerObject`; on the adversarial subset the rate is 16.9%. By comparison, the Stanford HAI 2024 study of LexisNexis LEXIS+ AI and Westlaw AI-ASSISTED RESEARCH reports hallucination rates of 17–33%. Our pipeline retrieves passages but the retrieved passages frequently do not lexically support the claim. We treat this as a strong signal that a cross-encoder reranker (e.g. `cross-encoder/ms-marco-MiniLM-L-12-v2`) is required before any production deployment, and we expose the metric prominently in the headline table rather than rolling it up.

**Per-product coverage gap (fairness).** Mondrian APS guarantees marginal coverage per group only when each group has sufficient calibration mass. Table 2 shows the "Debt or credit management" group at 0.778 against the 0.90 target – a 12-point under-protection on a vulnerable consumer category. The cause is small calibration support for that product after stratified sampling; a finite-sample correction or product-specific oversampling is needed before deployment. This pattern echoes regulator-facing concerns (e.g. the City National Bank \$31 M redlining consent order, 2023), which evaluate worst-subgroup behaviour rather than the marginal.

**Two-pronged abstention.** The reject option fires when $|S| \geq 3$ *or* $\hat{p}_{(1)} < \theta$; with mean set size $\approx$ 2.9 over six classes, this drives an abstention rate of 69.7% on the clean test set. Accuracy on the routed remainder is 89.6%. The take-away is that the system honestly admits "no single-route confidence" on most CFPB narratives – the conformal hedge is not actionable on a 5-minute SLA, but it is honest. A production deployment would route abstentions to a human queue rather than auto-decide.

**SCM confounders.** The backdoor adjustment over ($\textsc{Product}, \textsc{Issue}, \textsc{Conduct}$) omits institution, state, channel, jurisdiction, portfolio, and date. Real CFPB enforcement shows these matter: Wells Fargo's 2022 consent order (\$3.7 B redress; 16 M accounts) included Texas-specific GAP-refund failures affecting $\sim$90,000 borrowers; Bank of America's 2023 rewards issue varied by online vs. phone-application channel. Our ATEs should be read as conditional on these confounders being absent or randomly distributed, which they are not.

**Silver labels.** The baseline and the calibration target are silver routes derived from our rule+SBERT extractor; this avoids label leakage across systems but limits the absolute interpretation of macro-F1. A small human-labeled face-validity sample (we recommend 200–500 rows) is required before any external claim about routing performance.

**Strawman baseline.** TF–IDF+LogReg ties macro-F1 with the proposed system at 0.620 on silver labels. Published CFPB classification work reports $\sim$ 76.5% accuracy with deep encoders on 555,957 complaints; a DistilBERT or `intfloat/e5-small-v2` baseline would tighten the comparison.

**Frozen hypergraph.** The default encoder is one mean-pool pass over capped clique expansion; silhouettes are $\approx 0$ (conduct $-0.011$, product 0.010). The hypergraph manifold should be read as a viz-only diagnostic, not a learned representation. Switching `cfg.train_gat=True` runs a single contrastive epoch on (complaint, matching reg-section) positives.

**Single-jurisdiction corpus.** 51 sections cover the U.S. federal consumer-finance perimeter only; CRA/FHA, state UDAP statutes, and international regulators are out of scope.

**Stress-test power.** $n = 80$ per family is qualitative-probe scale; $n \geq 500$ is required for $p$-value-grade differentiation across families (cf. AdvGLUE, PromptBench).

**Reproducibility.** The committed `data/download_complaints.sh` pulls a 50,000-row sample directly from the CFPB public Consumer Complaint Database API; users no longer need access to the original `cfpb/data/processed/recent3y/` parquet shards.

**Deployment posture.** Given 6.1% clean citation faithfulness, a 12-point per-group coverage gap, and a 69.7% abstention rate, this system should be deployed only as decision support behind a human-in-the-loop queue, with mandatory escalation for fraud, foreclosure, debt-collection, and discrimination claims, and with audit logging on every routed decision. The CFPB's 60-day response SLA makes silent misrouting the dominant operational risk to mitigate.

## 9 Conclusion

We showed that a single SBERT encoder, a curated 51-section regulation corpus, and four classical components (regex+vector extraction, hybrid retrieval, post-hoc calibration plus Mondrian conformal sets, and a networkx SCM with backdoor adjustment) can be composed into a pipeline that surfaces *all four* of the properties commonly missing from production complaint classifiers: structure, grounded explanation, calibrated uncertainty, and an explicit causal lens. The complaint–regulation hypergraph manifold gives a single 2-D substrate on which to inspect both data sources jointly. We expect the framework to extend cleanly to other regulator-driven domains (SEC, FTC, OCC) by swapping the regulation corpus and retuning the conformal calibration set.

# A Schema

**ConductTypes (11):** unauthorized, deceptive_practice, unfair_fee, debt_collection_abuse, credit_report_error, loan_servicing_error, discrimination, dark_pattern, identity_failure, crypto_loss, ai_harm.
**HarmTypes (6):** monetary_loss, credit_damage, denial_of_service, privacy_violation, emotional_distress, discrimination_harm.
**RouteTargets (6):** fraud_ops, legal, supervisor, collections, servicing, route_to_other.
**SeverityLevel:** 1 (LOW) – 5 (CRITICAL).

**Default conduct $\rightarrow$ route mapping $\pi(\cdot)$ (used by the validator when no route is supplied):** unauthorized $\rightarrow$ fraud_ops; identity_failure $\rightarrow$ fraud_ops; crypto_loss $\rightarrow$ fraud_ops; deceptive_practice $\rightarrow$ legal; discrimination $\rightarrow$ legal; dark_pattern $\rightarrow$ legal; debt_collection_abuse $\rightarrow$ collections; loan_servicing_error $\rightarrow$ servicing; credit_report_error $\rightarrow$ servicing; unfair_fee $\rightarrow$ supervisor; ai_harm $\rightarrow$ supervisor.

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